

PRESTON (CANYANG) ZHAO

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Professional Summary

Current UW–Madison master’s student in computational data science with experience in software engineering, production ML systems, databases, and agentic systems, seeking a **Summer 2027 internship** in a software, data, or AI/ML development role.

Experience

Bio-Techne | Graduate Intern, Software Development

May 2026 – August 2026

Research analysis web platform

Minneapolis, MN

- Built an Azure-hosted TypeScript application and Python API for **50+ scientists**, replacing commercial analysis software with secure, configurable candidate filtering on internal research data; saved approximately **\$10K monthly** and cut analysis turnaround time by a third.
- Built an asynchronous Celery service with PostgreSQL, Docker, durable state, retries, and workload isolation; sustained **50+ completed jobs per workday at 99% job completion**, cutting median queue latency from **12 to 4 minutes**.
- Supported the application-side migration of **200 TB** from VM-local storage to ADLS Gen2; refactored data access and validated transfer integrity, enabling **10× execution capacity** and cutting job startup time by two-thirds.

Candidate-ranking system for R&D decision support

- Trained an **XGBoost ranker** in Databricks using structured candidate features, historical run outcomes, and scientist feedback; improved top-1 selection success from a **50% baseline to 70%** on held-out runs.
- Deployed a Dockerized **ML inference service** on Azure with batch prediction and an in-memory model; achieved **5-second median inference time** per batch and delivered ranked shortlists to scientists.

AI Rudder | LLM Engineer Intern

May 2025 – August 2025

Tool-calling language models for customer-service agents

Shanghai, China · On-site

- Fine-tuned Llama 3.1 8B with **LoRA SFT** on customer-service tool-call traces; reduced incorrect or missed calls from **12 to 9 per 100 eligible turns** when replaying 500 production conversations.
- Trained model candidates on examples of incorrect calls, missed calls, and invalid arguments; raised **strict tool-call correctness** from 61% with the base model to 68% with improved prompting and **80%** after fine-tuning, measured on the same held-out regression suite.
- Used a teacher LLM to generate paired call/no-call variations from labeled dialogs; aligned validation criteria with annotators and applied schema, argument, and duplicate checks to produce **150K+ validated conversations**.
- Built release evaluations combining tool-call checks with a three-level **LLM judge**; achieved **79% agreement with human judgments**, routed disagreements for human review, and recycled failures into training data.

Turning Green | Data Analyst Intern (Data Engineering)

December 2024 – March 2025

Data pipeline for farm-to-school site selection

Sausalito, CA · Hybrid

- Built **Airbyte-to-Snowflake ELT** for 18 months of CDFA, school, and market data; automated source loading and monthly warehouse refreshes, reducing manual data preparation from **four hours to 30 minutes** per update.
- Built **Snowflake SQL models** to reconcile county labels, school locations, and commodity categories; deduplicated records and validated joins, reducing the unmatched location-record rate from **12% to 2%** in farm-to-school coverage reports.
- Created a Snowflake feature pipeline for proximity analysis across **5K+ candidate routes**; trained an XGBoost viability model to rank hub sites, projecting **1.45×** the existing network’s San Mateo school coverage.

Projects

Mini-MapReduce | Big Data Systems Honors Project | Python, gRPC, Docker, HDFS, PyArrow

- Co-developed a Python **MapReduce runtime** and CLI using gRPC and Docker Compose; its scheduler dispatched tasks across workers while Hadoop HDFS and PyArrow/Parquet managed code, inputs, shuffle partitions, and outputs.
- Implemented worker registration and five-second heartbeats, 15-second liveness checks, and **task requeuing**; repaired stale gRPC channels after restarts and wrote **failure-injection tests** for worker termination, recovery, and post-restart jobs.

BadgerForge | Machine Learning Marathon Hack | Python, LangChain, LangGraph, Pydantic, Harbor, Docker

- Building an **autonomous coding agent** on a single GPU using LangChain and LangGraph to coordinate reasoning and shell tools; runs **Terminal-Bench** tasks in Docker through Harbor while tracking success and token usage.

Skills

Programming Languages: Python, C++, SQL, TypeScript, JavaScript

Cloud & Tools: Azure, AWS (S3, Redshift), Snowflake, Docker, Celery, Airbyte, dbt, PySpark, PostgreSQL, Databricks

AI/LLM: LoRA/SFT, LangChain, LangGraph, Agentic Tool Calling, Function-Call Evaluation, LLM-as-a-Judge, RAG

Analytics: XGBoost, Isolation Forest, EDA, Time-Series, Anomaly Detection, Feature Engineering

Education

University of Wisconsin–Madison

Expected Graduation: December 2027

M.S. in *Biomed Data Science* | Computer Science and Machine Learning Focus

B.S. in Data Science (Computational)

- Coursework:** Database Systems and Management, Big Data Systems, Applied Data Structures and Algorithms, AI Agents, Foundation Models
- Activities:** Graduate Teaching Assistant (Stats), Undergraduate Teaching Assistant (CS), Machine Learning + X Club